

Atlas Agent — Live on Casper Testnet

A production-ready walkthrough of Atlas running **live on Casper Testnet** (`casper-test`): an autonomous treasury agent that **buys evidence over the x402 payment protocol before it invests**, lets an LLM (Gemma 4 via OpenRouter) reason as the Treasurer, records **every** decision on-chain, and moves funds only through a vault that **enforces the treasury policy in WASM** — independently of anything the agent believes.

Everything below is real, on-chain state you can open yourself on `testnet.cspr.live`.

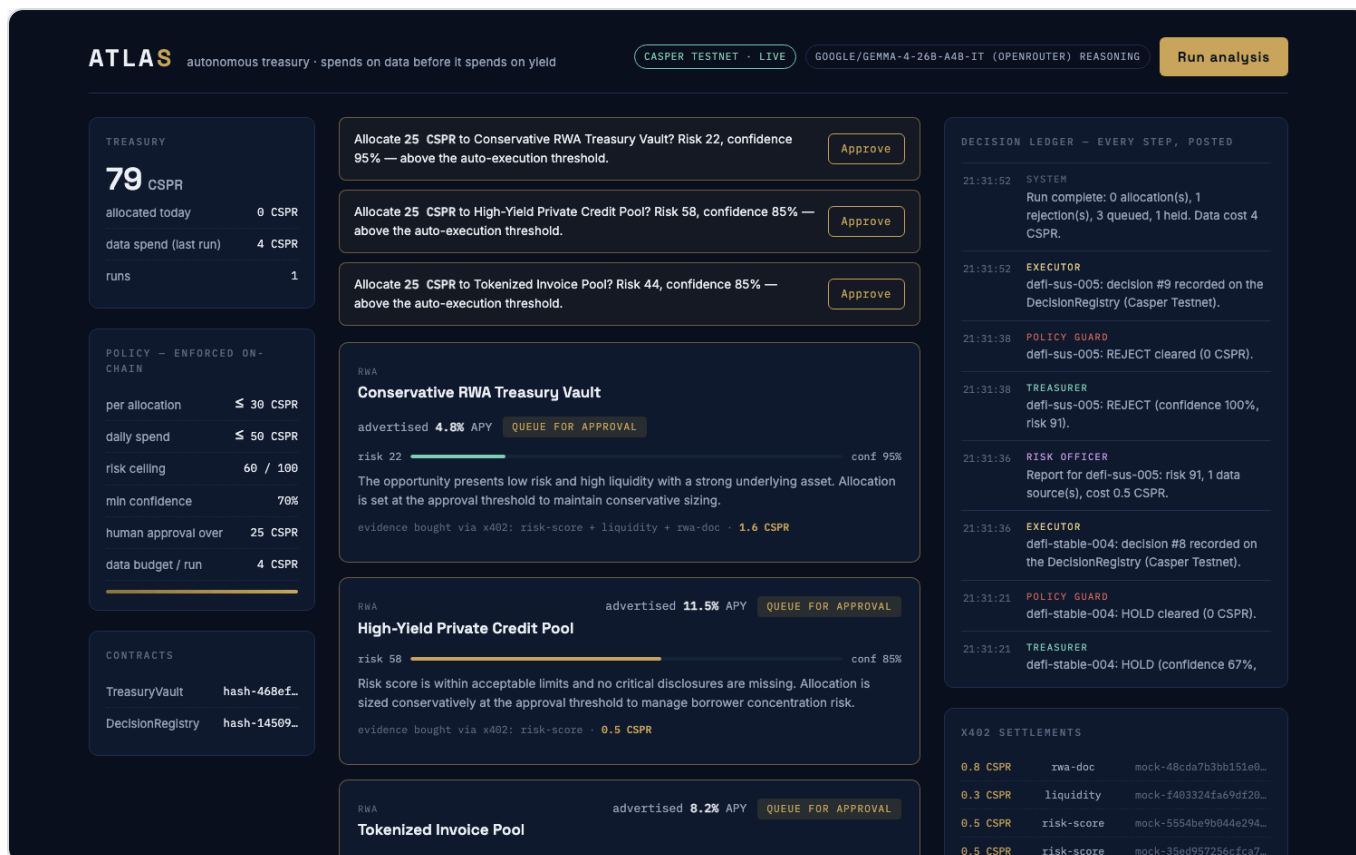
TL;DR — what's deployed

THING	VALUE
Network	Casper Testnet (<code>casper-test</code>)
TreasuryVault	<code>hash-468ef5d1...249d9c51</code>
DecisionRegistry	<code>hash-14509e44...1fd40cf8</code>
Agent / owner account	<code>017c500e...57d2c256</code>
Treasurer (LLM)	<code>google/gemma-4-26b-a4b-it</code> via OpenRouter
Vault deploy tx	<code>49440f21...2756fdb</code>
Registry deploy tx	<code>3506fba7...387650f</code>

On-chain vault state after the runs below: **balance 54 CSPR · 2 allocations executed**, with the policy `per-op ≤ 30 · daily ≤ 50 · min-confidence 70% · max-risk 60 · human-approval ≥ 25 CSPR`. The deployment runs the **two-key model** (separate agent + owner keys) with an **authenticated API**.

1. The live dashboard

The operator view at `http://localhost:3000`. The masthead reads **CASPER TESTNET · LIVE** and **GOOGLE/GEMMA-4-26B-A4B-IT (OPENROUTER) REASONING** — the agent is wired to the real contracts and an LLM Treasurer, not a simulation.



Three rails:

- **Treasury** (left) — live balance read from the on-chain vault (79 CSPR), data spend for the last run (4 CSPR), and the **policy that is enforced on-chain** (not in the prompt): per-allocation cap, daily cap, risk ceiling, min confidence, the human-approval threshold, and the per-run data budget. Below it, the two deployed contract hashes.
- **Opportunity desk** (center) — the marketplace, each card joined to the agent's latest decision with a colour-coded risk bar, the Treasurer's reasoning, and the **x402 evidence it paid for**.
- **Decision ledger** (right) — every pipeline step posted in real time, including **decision #9 recorded on the DecisionRegistry (Casper Testnet)**, plus the **x402 settlements** feed.

The full page, end to end:

TREASURY

79 CSPR

allocated today0 CSPR

data spend (last run)4 CSPR

runs1

POLICY — ENFORCED ON-CHAIN

per allocation≤ 30 CSPR

daily spend≤ 50 CSPR

risk ceiling60 / 100

min confidence70%

human approval over25 CSPR

data budget / run4 CSPR

CONTRACTS

TreasuryVaulthash-468ef...

DecisionRegistryhash-14509...

Allocate 25 CSPR to Conservative RWA Treasury Vault? Risk 22, confidence 95% — above the auto-execution threshold.

Approve

Allocate 25 CSPR to High-Yield Private Credit Pool? Risk 58, confidence 85% — above the auto-execution threshold.

Approve

Allocate 25 CSPR to Tokenized Invoice Pool? Risk 44, confidence 85% — above the auto-execution threshold.

Approve

RWA

Conservative RWA Treasury Vault

advertised 4.8% APY

QUEUE FOR APPROVAL

risk 22

conf 95%

The opportunity presents low risk and high liquidity with a strong underlying asset. Allocation is set at the approval threshold to maintain conservative sizing.

evidence bought via x402: risk-score + liquidity + rwa-doc · 1.6 CSPR

RWA

High-Yield Private Credit Pool

advertised 11.5% APY

QUEUE FOR APPROVAL

risk 58

conf 85%

Risk score is within acceptable limits and no critical disclosures are missing. Allocation is sized conservatively at the approval threshold to manage borrower concentration risk.

evidence bought via x402: risk-score · 0.5 CSPR

RWA

Tokenized Invoice Pool

advertised 8.2% APY

QUEUE FOR APPROVAL

risk 44

conf 85%

Risk score is within limits and no critical disclosures are missing. Allocation is sized conservatively at the approval threshold to manage risk.

evidence bought via x402: risk-score · 0.5 CSPR

DEFI

DeFi Stable Yield Pool

advertised 6.1% APY

HOLD

risk 35

conf 67%

Risk-adjusted score 2.9 is attractive but rests on a single data source; holding until corroborating data fits the budget.

evidence bought via x402: risk-score · 0.5 CSPR

DEFI

Quantum Yield Maximizer

advertised 94.0% APY

REJECT

risk 91

conf 100%

The risk score of 91 exceeds the maximum allowed threshold of 60 and the risk level is classified as critical due to unverifiable collateral and an anonymous team.

evidence bought via x402: risk-score · 0.5 CSPR

DECISION LEDGER — EVERY STEP, POSTED

21:31:52SYSTEM

Run complete: 0 allocation(s), 1 rejection(s), 3 queued, 1 held. Data cost 4 CSPR.

21:31:52EXECUTOR

defi-sus-005: decision #9 recorded on the DecisionRegistry (Casper Testnet).

21:31:38POLICY GUARD

defi-sus-005: REJECT cleared (0 CSPR).

21:31:38TREASURER

defi-sus-005: REJECT (confidence 100%, risk 0).

21:31:36RISK OFFICER

Report for defi-sus-005: risk 91, 1 data source(s), cost 0.5 CSPR.

21:31:36EXECUTOR

defi-stable-004: decision #8 recorded on the DecisionRegistry (Casper Testnet).

21:31:21POLICY GUARD

defi-stable-004: HOLD cleared (0 CSPR).

21:31:21TREASURER

defi-stable-004: HOLD (confidence 67%,

X402 SETTLEMENTS

0.8 CSPRrwa-docmock-48cda7b3bb151e0...

0.3 CSPRliquiditymock-f403324fa69df20...

0.5 CSPRrisk-scoremock-5554be9b044e294...

0.5 CSPRrisk-scoremock-35ed95725ecfca7...

0.5 CSPRrisk-scoremock-a18169bb8b20a4a...

0.5 CSPRrisk-scoremock-30fa3ecd79bbc64...

0.5 CSPRrisk-scoremock-495c959489bb47b...

0.4 CSPRmarket-signalmock-e8dc9fe283909f8...

2. What one run does — the six-role pipeline

Scout → Analyst → Risk Officer → Treasurer → Policy Guard → Executor
free endpoint pays per datum over x402, under a 4-CSPR budget composes a risk report per opportunity Gemma 4 / deterministic scorer (ALLOCATE/REJECT/HOLD) hard rules mirror the on-chain policy records EVERY decision on the DecisionRegistry, moves CSPR via the TreasuryVault

A live run on the demo marketplace produced these decisions (recorded on-chain as registry entries **#5 – #9**):

OPPORTUNITY	ADV. APY	RISK	TREASURER (GEMMA)	GUARD VERDICT
Conservative RWA Treasury Vault	4.8%	22	ALLOCATE 25 CSPR (conf 95%)	QUEUE_FOR_APPROVAL
High-Yield Private Credit Pool	11.5%	58	ALLOCATE 25 CSPR (conf 85%)	QUEUE_FOR_APPROVAL
Tokenized Invoice Pool	8.2%	44	ALLOCATE 25 CSPR (conf 85%)	QUEUE_FOR_APPROVAL
DeFi Stable Yield Pool	6.1%	35	HOLD	HOLD
Quantum Yield Maximizer	94.0%	91	REJECT (conf 100%)	REJECT

Two things to notice:

1. **The honeypot dies on paid evidence.** "Quantum Yield Maximizer" looks best on the free data (94% APY) but the agent *pays* for a risk screen and document analysis, learns the collateral is unverifiable and the team anonymous (risk 91), and rejects it — with the evidence shown on the card.
2. **The guard escalates large tickets.** Gemma proposed 25-CSPR allocations; since $25 \geq$ the human-approval threshold, the **Policy Guard converted every one to **QUEUE_FOR_APPROVAL**** — no funds move without a human. The vault enforces the same rule on-chain as defense in depth.

3. On-chain proof

Open the **TreasuryVault** contract on the explorer and its activity tells the whole story:

CSPR Products

CSPR live

Home

Blocks

Transactions

Contracts

Accounts

Validators

More

Wallet

8,311,429

LATEST BLOCK HEIGHT

5.91%

APY

\$0.0018

CSPR PRICE

11,321,389,251

CIRCULATING SUPPLY

Search by public key, hash, uref or CSPR.name

Q

Casper News:

Casper's multi-year roadmap for developers, institutions, and AI agents - [read more](#)

Contract Package

You're looking at the new "Contract Package" page. If you want to switch to old version, click [here](#).

468ef ... d9c51

General

Transactions

On-chain data

Contract info

Download Your Casper Wallet

Download Now

Activity

First

<

Page 1 of 1

>

Last

4c1d3 ... 5b978

1 hr ago · Block: 8,310,946 · Charge: 6.33822 CSPR

017c5 ... 2c256

execute_allocation with

468ef5d146cea36fa9ea25bf37806edf9bb41b420f336991edca78ce249d9c51

Transferred 21.00 CSPR (\$0.04) from uref-78187...594e9-007 to

017c5 ... 2c256

7e736 ... 33167

1 hr ago · Block: 8,310,926 · Charge: 1.48381 CSPR

017c5 ... 2c256

set_recipient with

468ef5d146cea36fa9ea25bf37806edf9bb41b420f336991edca78ce249d9c51

74df8 ... 73bb3

1 hr ago · Block: 8,310,925 · Charge: 7.15674 CSPR

017c5 ... 2c256

deposit with

468ef5d146cea36fa9ea25bf37806edf9bb41b420f336991edca78ce249d9c51

Transferred 100.00 CSPR (\$0.18) from 017c5 ... 2c256 to

uref-905f9 ... 375f5-007

Transferred 100.00 CSPR (\$0.18) from uref-905f9 ... 375f5-007 to

uref-78187 ... 594e9-007

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- **execute_allocation** — "Transferred 21.00 CSPR ... to 017c5...2c256 ". The agent actually moved treasury CSPR through the policy-checked vault entry point. (This 21-CSPR allocation came from an earlier run; with Gemma proposing 25-CSPR tickets, allocations now queue for human approval first.)
- **set_recipient** — the owner allowlisting the strategy recipient. A compromised agent **cannot** send funds to an arbitrary address; only owner-approved recipients are payable (see §5).
- **deposit** — the initial 100 CSPR funding of the vault.

Every decision the Treasurer made — ALLOCATE, REJECT, HOLD, or queued — is appended to the **DecisionRegistry**, an on-chain, append-only audit trail: [open the registry contract](#) → (look for the **record_decision** calls). The agent's **account activity** shows the full sequence of deploys, **record_decision**, **execute_allocation** and **set_recipient** transactions.

4. The Treasurer: Gemma 4 via OpenRouter

The Treasurer is pluggable. The reasoning engine runs a **provider cascade**:

```
OpenRouter (Gemma 4) → Claude → deterministic scorer
(if OPENROUTER_API_KEY) (if ANTHROPIC_API_KEY) (always available)
```

- The model receives the risk report + policy and must answer in **strict JSON**, schema-validated with zod; raw control characters are sanitized so smaller models parse reliably.
- If a provider errors, the pipeline **falls through** to the next — the treasury never stalls on a bad model response. (In a live run, one Gemma reply with a stray newline fell through to the deterministic scorer mid-run; the run completed and all five decisions still landed on-chain.)
- The LLM only ever **recommends**. The Policy Guard and the on-chain vault decide what executes.

Switch models with one env var (`OPENROUTER_MODEL`), including the free `:free` Gemma variants.

5. Why this is safe — the security model

The whole premise is that the **vault enforces the mandate on-chain**, so a fully compromised agent still cannot exceed its limits. An adversarial review hardened the contract before deploy:

- **Recipient allowlist** — agent-initiated allocations may only pay owner-approved addresses (`set_recipient`). This is the primary guard against an agent draining funds to an attacker.
- **Cumulative per-opportunity cap** — `allocated[opp] + amount ≤ max_allocation_per_op` over the opportunity's lifetime, so the cap can't be bypassed by chunking.
- **Per-call human-approval threshold, daily cap, risk ceiling, confidence floor, pause** — all checked inside `execute_allocation`, all owner-governed (`set_policy` / `set_agent` / `set_paused` / `withdraw` are owner-only).
- **Append-only audit** — the DecisionRegistry has no update/delete path; `timestamp` and `recorded_by` are stamped by the contract.
- **Two-key separation (live)** — a dedicated **agent key** signs the agent's autonomous calls (record-decision + sub-threshold execute-allocation); a separate **owner key** signs human approvals. So the recipient allowlist and the approval gate actually *bind* to the agent.
- **Authenticated API** — `POST /api/run` and `POST /api/approve` require a bearer token (`AGENT_API_TOKEN`); read-only endpoints stay open for the dashboard.

The contract ships with **12 passing unit tests** covering each of these, including the recipient allowlist and the cumulative-cap chunking defense.

Verified on-chain. The agent account signs and pays gas for the decision records; an agent-key attempt to move $\geq$ the approval threshold reverts `RequiresHumanApproval`, while the owner key succeeds via the approve endpoint. The cumulative per-op cap also reverted a real over-allocation to an already-funded opportunity (`ExceedsPerOpportunityLimit`).

6. Reproduce it yourself

```
# 1. Build & test the contracts (nightly toolchain is pinned in rust-toolchain.toml)
cd contracts
cargo test                                # 12 unit tests
./build-wasm.sh                           # per-module wasm artifacts
cargo build --release --features livenet --bin atlas_livenet

# 2. Deploy to Casper Testnet (needs a funded key; see .env.example for the 4 ODRA_* vars)
export ODRA_CASPER_LIVENET_NODE_ADDRESS=https://node.testnet.casper.network/rpc
export ODRA_CASPER_LIVENET_EVENTS_URL=https://node.testnet.casper.network/events
export ODRA_CASPER_LIVENET_CHAIN_NAME=casper-test
export ODRA_CASPER_LIVENET_SECRET_KEY_PATH=/path/to/secret_key.pem
./target/release/atlas_livenet deploy      # prints vault + registry hashes
./target/release/atlas_livenet fund-vault <vault> 100000000000
./target/release/atlas_livenet set-recipient <vault> <recipient> true

# 3. Run live. In .env: DRY_RUN=false, contract addresses, agent + owner key paths,
#   AGENT_API_TOKEN, and OPENROUTER_API_KEY (Gemma). See .env.example.
set -a; . ./env; set +a
npm run start -w services                 # x402 data marketplace on :4021
npm run serve -w agent                    # agent API on :4030 (LIVE)
npm run web                               # dashboard on :3000
```

Open the dashboard, press **Run analysis**, and watch Gemma reason, the guard escalate, and the decisions land on `testnet.cspr.live`.

Built for the Casper Agentic Buildathon 2026. Money only moves where evidence was bought — and only within a mandate the chain enforces.